Psoriasis Treatments and Comorbidities

Metabolic Syndrome and Cardiovascular Risk

Systemic treatments for psoriasis are valuable not only for controlling skin symptoms but also for potentially improving aspects of metabolic syndrome, which is commonly associated with psoriasis. However, the impact of these treatments on metabolic and cardiovascular risk factors varies.

There is ongoing debate about whether newer therapies, particularly TNF inhibitors, lead to meaningful improvements in physiological markers of metabolic syndrome. Some studies suggest potential benefits, while others show neutral or even adverse effects. For example, weight gain has been observed in some patients using TNF inhibitors.[37]

Specific concerns include: - **Cyclosporine**: May worsen atherogenic dyslipidaemia, arterial hypertension, and glucose intolerance. - **Acitretin**: Can exacerbate atherogenic dyslipidaemia. - **Methotrexate**: Requires caution in patients with increased risk of liver or renal toxicity.[38]

Therefore, careful monitoring and follow-up are recommended for patients receiving these therapies.

We also recommend increased surveillance for metabolic syndrome in psoriasis patients, including assessment of: - Increasing waist circumference - Elevated blood pressure - Raised triglyceride levels - Reduced HDL cholesterol - Elevated fasting blood glucose

Additionally, patients should receive appropriate lifestyle interventions such as obesity management and smoking cessation counseling when indicated.

Type II Diabetes and Insulin Resistance

From a safety standpoint, none of the available biological or non-biological systemic treatments for psoriasis are contraindicated in patients with type 2 diabetes or insulin resistance. However, clinical decision-making is limited by a lack of comparative data.

Some evidence suggests that anti-TNF agents—such as **infliximab**, **adalimumab**, **certolizumab pegol**, and **etanercept**—may reduce insulin resistance.[39–41] However, one study found no additional benefit of combining anti-TNF therapy with methotrexate over methotrexate alone in terms of HbA1c or fasting glucose levels.[42]

Other agents, including **methotrexate** and **acitretin**, have also been associated with improved insulin sensitivity in some studies.[43]

Caution is advised when using **cyclosporine** and **methotrexate** in diabetic patients due to increased risks of hepatotoxicity and nephrotoxicity.[38]

All systemic treatments currently available can be used in psoriasis patients with type 2 diabetes and/or insulin resistance. Recommended options include: - **TNF antagonists**: adalimumab, certolizumab pegol, etanercept, infliximab - **IL-12/23 inhibitor**: ustekinumab - **IL-23/p19 inhibitors**: guselkumab, risankizumab, tildrakizumab - **IL-17 receptor blocker**: brodalumab - **IL-17 inhibitors**: ixekizumab, secukinumab - **Synthetic drug**: apremilast - **Conventional drugs**: fumarates, acitretin

Cyclosporine and methotrexate may also be used, but with closer monitoring.

Obesity in Psoriasis Patients

For obese patients with psoriasis, the following systemic treatments are recommended: - **TNF antagonists**: adalimumab, certolizumab pegol, etanercept, infliximab - **IL-12/23 inhibitor**: ustekinumab - **IL-23/p19 inhibitors**: guselkumab, risankizumab, tildrakizumab - **IL-17 receptor blocker**: brodalumab - **IL-17 inhibitors**: ixekizumab, secukinumab - **Synthetic drug**: apremilast - **Conventional drugs**: fumarates, acitretin

Dietary interventions are encouraged, as they have been shown to improve outcomes with biologic therapy.[44]

From an efficacy perspective, studies indicate that certain biologics—**ustekinumab**, **infliximab**, **adalimumab**, and **etanercept**—as well as most conventional therapies, may require higher dosing in obese patients compared to those with healthy body weight.

Weight-adjusted dosing may therefore be necessary for optimal response in obese individuals. Currently, only **infliximab** and **ustekinumab** have established weight-based dosing regimens.[45,46] The need for dose adjustment with newer biologics remains less clear and requires further study.